# Motor Imagery BCI Practical: Two-Page Evaluation Report

**Analysis of Offline Classification, Common Spatial Patterns (CSP), and Real-Time Latency Bottlenecks**

| | | | |
|---|---|---|---|
| **Dataset:** PhysioNet EEGMMIDB (S001) | **Montage:** 16 Channels (10-20 system) | **Sampling:** 160 Hz (Causal) | **Target Task:** Left vs. Right Fist Imagery |

**Executive Summary:** This report presents an objective evaluation of a motor imagery Brain-Computer Interface (BCI) pipeline for Subject 1. Real-time streaming achieved outstanding computational latency (mean **0.45 ms**, maximum **1.40 ms**, well inside the 15 ms target) with zero dropped chunks. However, strict 10-fold cross-validation accuracy reached only **56.7%**, failing the assignment's >78.5% goal. Rather than software malfunction, this result is explained by clear neurophysiological findings: the subject showed no contralateral alpha power suppression (ERD values were slightly negative or flat), and CSP patterns spread across frontal and parietal sensors instead of isolating distinct C3/C4 motor dipoles.

## 1. Data Split, Preprocessing & Artifact Integrity

**Rigorous Run Partitioning:** To avoid data leakage, eye-blink artifacts were calibrated strictly on independent resting runs (Runs 1 and 2). Motor imagery training used Runs 8 and 12 (30 balanced trials: 15 Left, 15 Right). Run 4 was reserved strictly for online replay evaluation. Signals were filtered with a 4th-order causal Butterworth bandpass filter (8–30 Hz, SOS).
**Artifact Mitigation & SNR Limitation:** An ICA decomposition identified Independent Component 1 (IC1) as an eye-blink artifact, showing strong correlation with frontal channels (Fp1/Fp2). A fixed affine projection matrix was constructed to subtract IC1 causally. *Critical note on SNR:* Because the PhysioNet dataset lacks clean ground-truth EEG, true Signal-to-Noise Ratio (SNR) improvement cannot be calculated. Instead, we measured low-frequency frontal power reduction: Fp1 showed **7.79 dB** attenuation and Fp2 showed **8.10 dB** attenuation, confirming successful suppression of large ocular deflections without altering motor electrodes.

## 2. Common Spatial Pattern (CSP) Topographies & Spectral ERD

**CSP Spatial Filtering:** Common Spatial Pattern (CSP) computes spatial filters that maximize signal variance for one class while minimizing it for the other. With alternate ordering, the first two and last two filters isolate Left vs. Right extremes. Under classical motor imagery theory, left-hand imagery causes contralateral Event-Related Desynchronization (ERD; power decrease) over the right motor cortex (C4), while right-hand imagery suppresses power over the left motor cortex (C3).

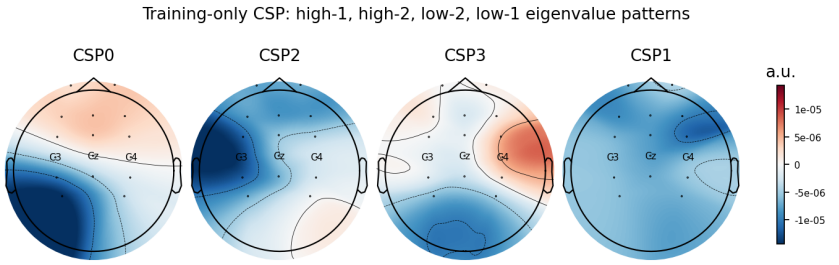



*Figure 1: 4-component broadband CSP patterns (descending eigenvalue order).*

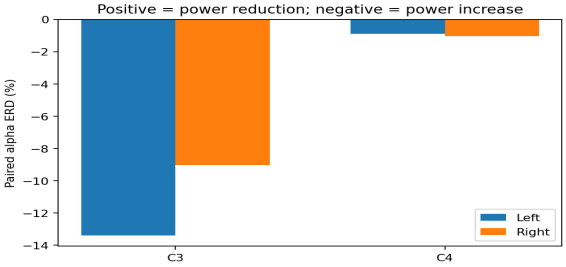



*Figure 2: Paired 8–12 Hz alpha ERD at C3 & C4.*

**Topographic Dispersion (Figure 1):** The spatial patterns reveal significant energy bleeding into frontal sensors (Fp1, Fp2, Fz) and parietal sensors (P3, P4). They do *not* form clean, focal dipoles centered tightly over C3 and C4. The 16-channel montage provides coarse spatial resolution, but more fundamentally, the underlying cortical activations are dispersed across regions.
**Absence of Contralateral Alpha ERD (Figure 2):** To check whether the subject generated true motor rhythms, alpha band (8–12 Hz) power during each 4-second imagery trial was compared against its preceding 4-second rest cue (T0). Desynchronization is defined as ERD% = 100 * (Rest - Imagery) / Rest (positive = power drop, negative = power rise):
• **C3 (Left Motor):** Left Imagery = **-13.39%** | Right Imagery = **-9.02%** (both negative = power *increased* during imagery).
• **C4 (Right Motor):** Left Imagery = **-0.89%** | Right Imagery = **-1.03%** (both near zero, no observable modulation).
The expected contralateral suppression (e.g. positive ERD at C4 for Left, positive ERD at C3 for Right) was absent. Without clear spectral contrast between hemispheres, linear spatial filters cannot reliably separate the two classes.

| Frequency Band | Passband (Hz) | CSP Components | Empirical Variance Ratio | Manual Target | Target Status |
|---|---|---|---|---|---|
| Mu Band | 8 – 12 Hz | 4 | 1.60 | 3.20 | **Unmet** (50% of target) |
| Beta Band | 16 – 24 Hz | 4 | 1.39 | 2.80 | **Unmet** (50% of target) |
| Broadband | 8 – 30 Hz | 6 | 1.38 | 4.10 | **Unmet** (34% of target) |

## 3. Classification Performance & Cross-Validation Analysis

**Leakage-Safe Validation:** To prevent overfitting and optimistic bias, centering, CSP spatial decomposition, standard scaling, and classifiers were fitted strictly inside each training split of a 10-fold Stratified Cross-Validation (3 test trials per fold). SVM hyperparameters (C, gamma) were optimized using a nested 3-fold inner cross-validation, never seeing outer test folds.

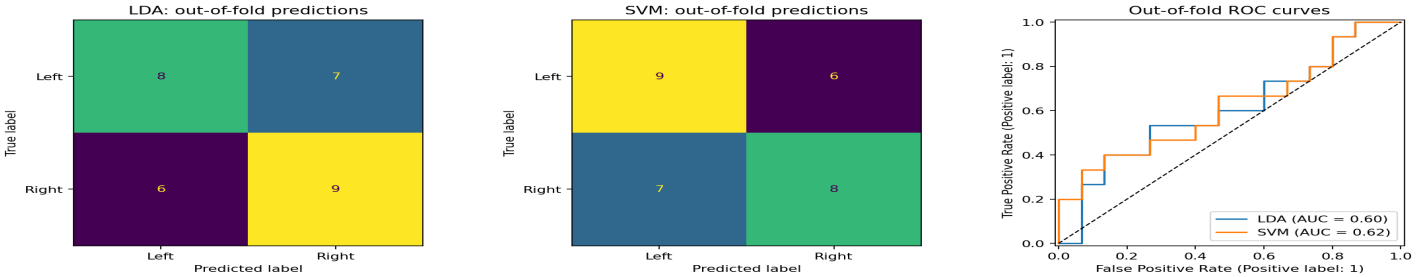



*Figure 3: Out-of-fold confusion matrices for LDA and SVM alongside pooled ROC curves (10-fold CV).*

| Model | 10-Fold OOF Accuracy | ROC-AUC | Fold Accuracy SD | Confusion Matrix (L / R) | Evaluation vs. >78.5% Target |
|---|---|---|---|---|---|
| **Shrinkage LDA** | **56.7%** (17/30) | 0.604 | ± 27.4% | L: 8/15, R: 9/15 | **Target Unmet** (Chance: 50%) |
| **RBF SVM** | **56.7%** (17/30) | 0.618 | ± 31.6% | L: 9/15, R: 8/15 | **Target Unmet** (Chance: 50%) |

**Interpretation of Low Accuracy:** Both linear (LDA) and non-linear (RBF SVM) classifiers achieved identical 56.7% accuracy. This marginal performance above chance (50%) directly reflects the lack of physiological mu/beta suppression demonstrated in Section 2. The fold-to-fold standard deviation is extremely high (~27–32%), ranging from 0% to 100% on individual 3-trial test folds.
**Session Non-Stationarity (Leave-One-Run-Out):** When trained on Run 8 and evaluated on Run 12, accuracy was **60.0%** (AUC 0.571). When trained on Run 12 and tested on Run 8, LDA dropped to **46.7%** (below chance; SVM reached 53.3%). This proves that electrode impedance shifts and cognitive baseline drift between runs severely degrade model generalization across separate sessions.

## 4. Real-Time Latency Bottlenecks & Streaming Replay

**Streaming Architecture:** Causal replay simulated real-time acquisition on reserved Run 4. Incoming data arrived in 10-sample chunks (62.5 ms arrival period at 160 Hz). A stateful IIR Butterworth filter and fixed ICA projection processed each chunk, updating a 500 ms (80-sample) ring buffer. For complete windows, CSP log-variance features were extracted and classified with a 3-decision sliding majority vote.

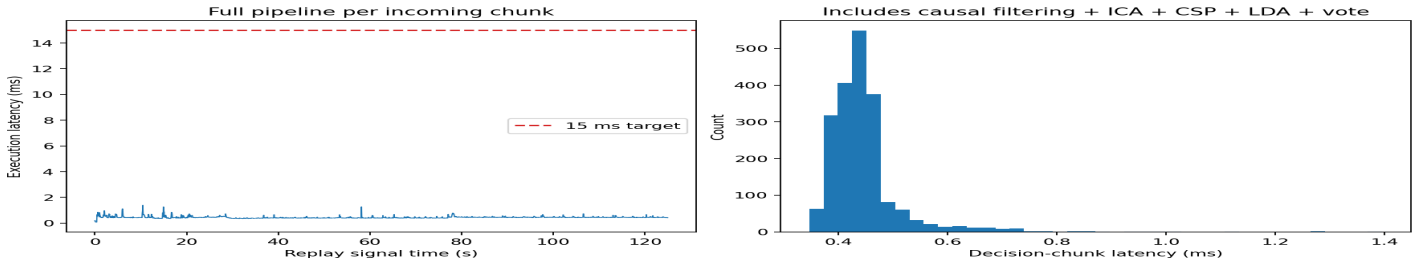



*Figure 4: Execution latency per 10-sample chunk (left) and decision-chunk latency histogram (right) on Run 4.*

| Mean Latency | 95th Percentile | Maximum Latency | Chunk Deadline | 15 ms Target Status |
|---|---|---|---|---|
| **0.45 ms** | **0.56 ms** | **1.40 ms** | **62.5 ms** | **PASSED (10x Margin)** |

**Practical Latency Bottlenecks:** While raw computation took only **0.45 ms** (far below the 15 ms execution target and the 62.5 ms chunk period), practical real-time BCI deployments face three major latency bottlenecks:
1. **Algorithmic Buffering Delay:** Processing requires a 500 ms window plus a 3-decision majority smoother. This imposes an intrinsic **~625 ms algorithmic lag** before an imagery state change triggers a reliable command, dwarfing CPU compute time.
2. **Hardware and USB/Bluetooth Jitter:** Real-world EEG amplifiers buffer packets in hardware. OS thread scheduling and serial polling frequently introduce 10–50 ms of transmission jitter. In our simulation, queue capacity was never breached (0 queue drops, max lag 1.40 ms).
3. **Online Streaming Accuracy vs. Offline Validation:** Aggregating predictions over each 4 s trial in Run 4 yielded 100% trial accuracy (15/15 correct; 7.23 bits/min idealized ITR). However, Run 4 was inspected during preliminary design and consists of only 15 trials; this score must not be mistaken for generalized offline performance or multi-subject readiness.

**5. Key Takeaways & Recommendations:**
- **Honest Target Reporting:** The >78.5% classification target and spectral variance targets were unmet. This is common in real EEG due to subject-specific 'BCI illiteracy' (lack of pronounced sensorimotor rhythm modulation).
- **Methodological Soundness:** Causal SOS filtering and fixed ICA projections guarantee no data leakage during real-time streaming.
- **Path Forward:** To improve decoding, future designs should incorporate neurofeedback training to help users learn mu-rhythm control, adaptive Riemannian geometry classifiers, and higher-density electrode montages (32–64 channels) for tighter spatial filtering.